Prithwijit Ghosh

Data Scientist Senior Analyst | Accenture Technology

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Data Scientist at **Accenture** with **2.5+** years of experience and a Master's from **IIT Kanpur**. Delivered analytics solutions to **six+** global clients across sales, finance and operations. Built production-grade models for accounts receivable forecasting and late payment prediction, cash flow, and sales forecasting. Skilled at translating insights into executive-facing Excel and Power BI dashboards and communicating effectively with leadership. Collaborative team player with a strong focus on delivering high-quality client solutions.

Professional Experience *July 2023 – Present*

Sales and Guest-Count Forecasting

- Developed an ensemble forecasting model for ***sales*** and ***guest-count***, projecting **48 months** ahead for business planning.
- Integrated macroeconomic indicators such as ***unemployment rate*** and ***price index*** to enhance precision.
- Implemented multiple models including **Prophet**, Theta, MSTL, LGBM, and Naive, totaling over **450** forecasts
- Designed an ***evaluation vector*** incorporating **trend** and **seasonal** MAPE for a unique performance assessment.
- Optimized horizon-based model selection through minimization of the **Euclidean norm** for reliability.
- Classified the models into ***overpredictive*** and ***underpredictive*** categories to enhance the power of ensembling.
- Incorporated pre- and post-processing techniques with different ensemble metrics to balance out the fluctuations.
- Applied exponentially weighted ***smoothing*** over six months to stabilize forecast fluctuations for longer horizons.
- Delivered accurate forecasts achieving **1–3%** MAPE for 24 months and **2–5%** for 48 months for both targets.

Intelligent Collections 3.0

- **Accounts Receivable (AR) Forecasting**
  - Focused on short-term account receivable forecasts, typically 1 month and up to 6 months for business planning.
  - Designed models across client segments including SMA, EWMA and business-specific forecast filters.
  - Implemented advanced time series models like ARIMA, SARIMA, **Theta** and **MSTL** for pattern recognition.
  - Developed ML and DL models such as XGBoost, LightGBM, CatBoost and ***Temporal Convolutional Network***
  - Achieved forecast errors within **2–5%** across all AR buckets — *Not Yet Due, Current Due* and *Over Due*.
  - Built and deployed an interactive ***Power BI*** dashboard to present findings effectively to leadership and client demo.
- **Late Payment**
  - Conducted deep customer profiling and entity analysis to predict future late payers with higher precision.
  - Trained ***XGBoost*** models using top *50 features* to predict late chance and duration at due date and due month levels.
  - Achieved AUC-ROC of **84%** for due-date late and **90%** for due-month late, with **5-day** MAPE for late days.
  - Developed a ***risk score*** and ***risk categorization*** combining all the predictions with a ***quantile calibration***.
  - Delivered **38%** reduction in *overdue amounts*, **15%** lower **open AR** and **12%** higher *collections*.

Cash-Flow Forecasting

- Delivered end-to-end B2C cash flow forecasting with ***daily, weekly*** and ***monthly*** projections for 12 months.
- Processed 90 GB (500 million rows) of raw data from **6 million** active customers using **Polars** for segmentation.
- Built meta-learning models that improved Cash-In forecast accuracy to **98%** (↑ from 70%) using boosting approach.
- Improved Cash-Out forecast accuracy to **93%** (↑ from 64%) through adaptive time series and ML model stacking.
- Consolidated the complete analysis into an interactive fully functional ***Power BI*** dashboard for client presentations.

MSc Final Semester Project *May 2023 – July 2023*

Classification using Robust DD Classifier for Cellwise Contaminated Outliers | *Mentor - Dr. Subhajit Dutta*

- Identified and estimated ***cellwise outliers*** from multivariate data using 2SGS and MCD estimator (rather than casewise).
- Transformed data into a ***k-dimensional space*** for k-class classification problem while preserving class separation.
- Evaluated multiple ML models, including ***Logistic Regression, KNN, Random Forest*** and ***SVM*** with various kernels.
- Proposed a ***Robust 2SGS classifier*** outperforming benchmarks across simulated cases *(exp, non-exp and mixture family)*
- Achieved moderate to decent improvements on real-life benchmark datasets that had cellwise outliers.

Skills

- ***Programming Languages:*** Python, SQL, R, LaTeX, Bash-Scripting
- ***Software:*** MS Excel, MS PowerPoint, Minitab, Maple
- ***Deployment Tools:*** Power BI, Streamlit, AWS, Git
- ***Libraries:*** Numpy, Pandas, Polars, DuckDB, Matplotlib, Scikit-learn, XGBoost, Darts, Tensorflow, Keras
- ***Domains:*** Data Science, Data Analysis, Predictive Modelling, Statistics, Time Series, Inference, Business Intelligence

Education

M.Sc in Statistics || Indian Institute of Technology, Kanpur (**IITK**) || ***CGPA:*** *8.9* — *August 2021 - July 2023*

* *A*/A :* 7 courses (Multivariate analysis, Simulation, Bayesian Analysis)
* *Final Project:* Robust Depth-Depth Classifier for Cell-wise Contaminated Outliers
* Received grade A (**highest**) in the M.Sc. final project.

B.Sc in Statistics || Bidhannagar College || ***CGPA:*** *9.99* — June 2018 - July 2021

* *A*/A:* All courses (total 26)
* Secured All India Rank **(AIR) 7** in Joint Admission test for Masters **(JAM)** on Mathematical Statistics (MS).
* Selected to participate in **MTUSS (2019)**, one of the most prestigious mathematics workshops, organized by *National Board of Higher Mathematics* ***(NBHM)***.

Certifications

Data Analysis with R (Google) *Machine Learning Specialization (Stanford University)* *Python 101 for Data Science (IBM)*